Project proposal

Team 12 - Alon Bar, Amir Belinsky, Tal Ber and Nitzan Graf

Section 1 - Introduction

LLMs are now commonplace and are widely used by individuals and companies, while model prices range from a few cents to hundreds of dollars per million tokens. We were curious if we are actually getting good value for our money. Existing platforms usually compare models using overall price versus performance scores, but they treat intelligence as one combined capability.

Our research asks a more specific question: does $1 buy you more coding skill, scientific reasoning, or general knowledge in today's LLM market? In other words, is the market pricing LLMs as one big "intelligence" bundle, or are some skills cheaper to buy than others?

To answer this, we compare model cost efficiency across several benchmark categories, including coding , advanced reasoning , science , and general knowledge. Rather than looking only at raw benchmark scores, we analyze price per performance point in each domain.

This question matters because different applications require different skills. A company building a coding assistant may care more about coding performance than general reasoning. If some capabilities are consistently cheaper for example, coding from open source models while frontier reasoning remains expensive, organizations can make more cost effective choices.

Our approach uses a March 2026 snapshot of LLM prices and benchmark scores. We fit separate price versus performance curves for each benchmark and compare the results to see whether different skills follow different market pricing patterns

Our hypotheses are: (H1) the price–performance slope differs across the five benchmarks, with coding cheapest per score point and frontier reasoning (humanitys_last_exam) most expensive; (H2) open-source models sit systematically below the price–performance line of closed-source models, controlling for score.

Section 2 - Data

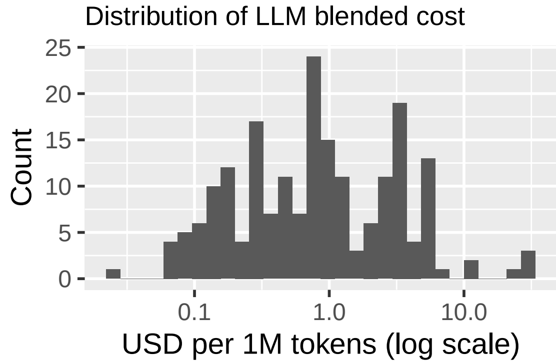
Source: Kaggle - LLM Price-Performance Tracker (March 2026) by kanchana1990. https://www.kaggle.com/datasets/kanchana1990/llm-price-performance-tracker-march-2026

Each row is one LLM model as observed in a single snapshot taken on 2026-03-31. The variables we use:

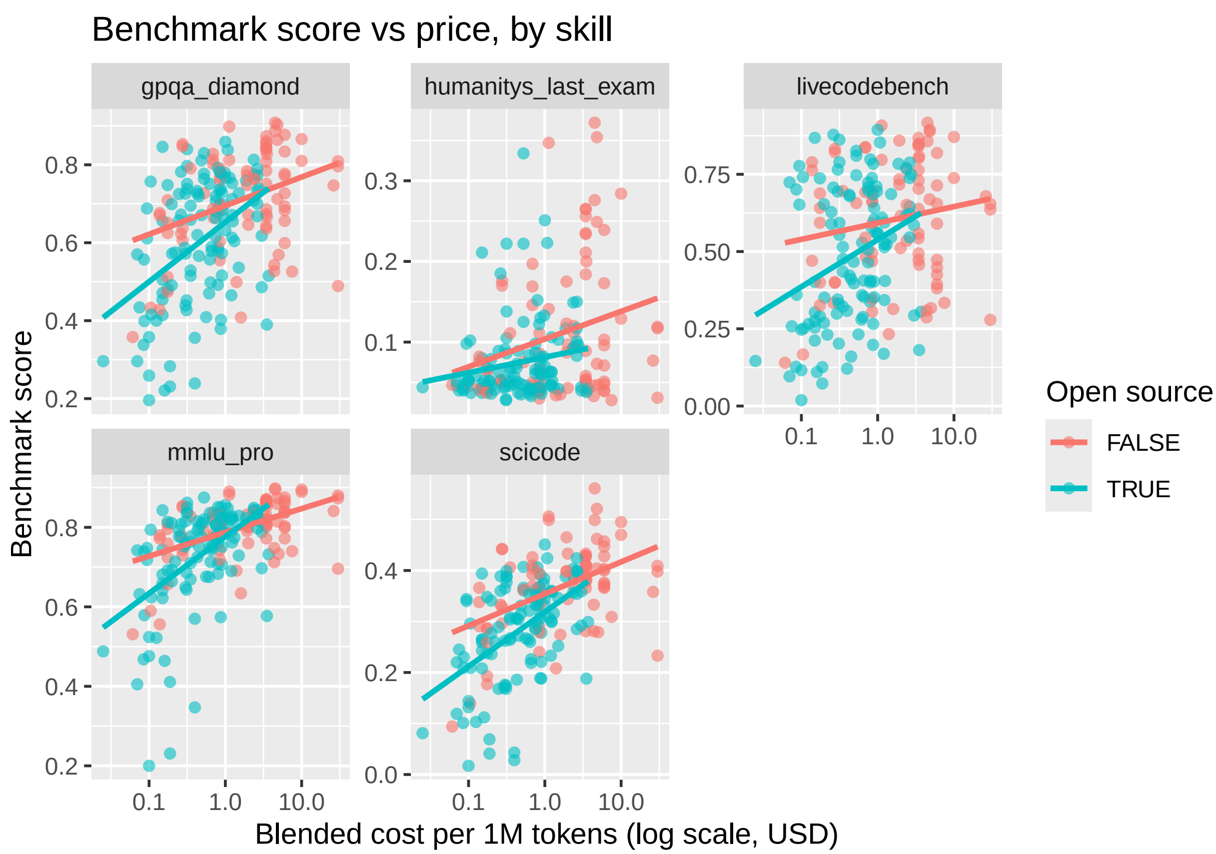
- Identifiers: model_name, provider
- Overall scores: aa_intelligence_index, aa_coding_index, composite_benchmark
- Per-skill benchmarks: mmlu_pro (general knowledge), gpqa_diamond (science), humanitys_last_exam (multidisciplinary performance), livecodebench (coding), scicode (scientific coding)
- Categories: pricing_tier, is_open_source
- Prices in USD per 1M tokens: input_cost_usd_per_1m, output_cost_usd_per_1m, blended_cost_usd_per_1m
- Discrete numeric: release_year and parameter_count (in billions). The mix of categorical (provider, pricing_tier, is_open_source), discrete numeric (release_year, parameter_count), and continuous numeric variables (benchmark scores, prices) fits the project requirement of having varied types.

Section 3 - Preliminary results

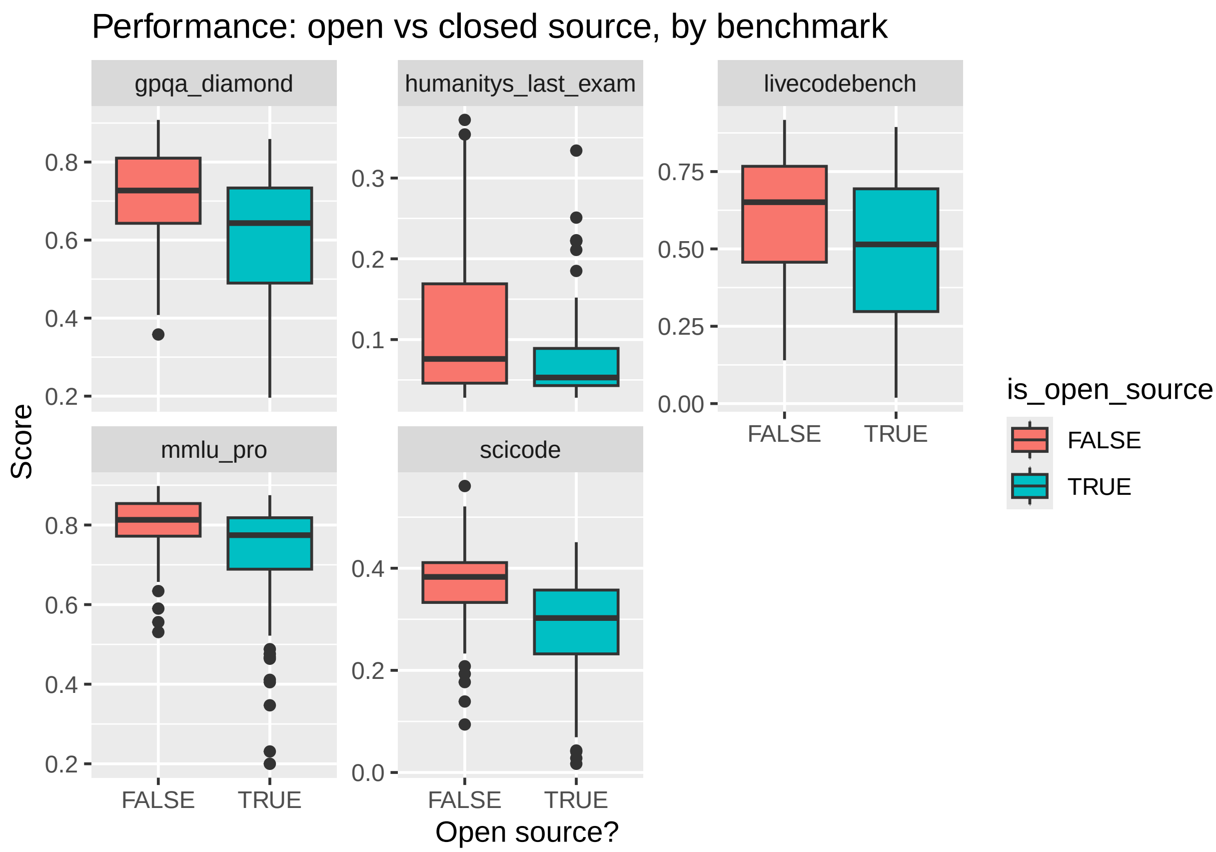
Price distribution. Prices span several orders of magnitude, so we work on a log scale.



Price vs each benchmark, side by side. This is the core picture of our question - one panel per skill, same axes, so we can compare slopes directly.



Open vs closed source - performance gap by benchmark.



Summary table.

```
## # A tibble: 25 x 5
##    provider       n mean_intel mean_coding mean_price
##    <chr>      <int>      <dbl>       <dbl>      <dbl>
##  1 KwaiKAT        1       36          18.3      0.525
##  2 Xiaomi         2       34.8        28.8      0.15
##  3 Anthropic     15       34.1        31.7     10.5
##  4 MiniMax        3       33.3        25.5      0.671
##  5 Kimi           3       32.7        27.6      1.07
##  6 xAI            8       31.0        25.8      1.75
##  7 OpenAI        33       31.0        27.3      3.06
##  8 DeepSeek      13       27.2        26.8      0.913
##  9 Google        13       26.9        23.5      1.34
## 10 Z AI          10       25.7        23.1      0.789
## # i 15 more rows
```

What we expect to see: a positive but noisy link between price and performance for every benchmark, slopes that differ across skills, open-source models clustered at lower price, and a handful of providers dominating the high-price / high-score corner.

Section 4 - Work plan

Although our question is phrased as 'what does $1 buy', we model log(price) as the response and benchmark scores as predictors. This is the standard hedonic-pricing framing: the regression coefficient on a benchmark estimates how much extra the market charges for one more unit of that skill, holding others fixed, which is the inverse of (and equivalent to) 'price per score point'.

Outcome (Y): blended_cost_usd_per_1m, but we take its log() to make varying values more manageable. Predictors (X): the five skill benchmarks: - mmlu_pro - knowledge score - gpqa_diamond - science reasoning score - humanitys_last_exam - multidisciplinary performance score - livecodebench - real-world coding score - scicode - scientific coding score plus is_open_source and provider as controls.

Comparison groups: open-source vs. closed-source models (a true between-group comparison). The five benchmarks are not comparison groups but five separate outcomes measured on the same models; we compare the coefficients across the five benchmark regressions, not the models across benchmarks.

Methods: - One simple linear regression of log(price) on each benchmark, then compare the slopes - this gives us the percentage price premium per score point for each skill. - One multiple regression of all benchmarks together, to isolate how each skill actually effects prices on its own. - For each benchmark regression we extract the residuals. A negative residual identifies a model that is cheaper than its score predicts. We then compute the pairwise Spearman correlation of residuals across the five regressions to test whether the same models are consistently cheap across multiple skills. - To compare open-source and closed-source models, we run a Welch's two-sample t-test on the residuals per benchmark, and we add is_open_source as a predictor in the multiple regression to test whether open-source models are systematically cheaper after controlling for skill.

We are splitting tasks into parts and working on it at the same time, where each of us working on a different topic like: cleaning the data, making the graph, writing and documenting the process. We also switch the work sometimes so its not a constant one person per topic.

Appendix

Data README

```
# SISE2601 Project data description
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Team 12 - Alon Bar, Amir Belinsky, Tal Ber and Nitzan Graf

# Project Data Description
Data File: llm_price_performance_tracker_2026-03-31.csv
Data source: https://www.kaggle.com/datasets/kanchana1990/llm-price-performance-tracker-march-2026

Total Observations (Rows): 209
Total Variables (Columns): 19
## Detailed Variable

model_name (Text): The official commercial or research name of the model.
model_slug (Text): A URL, lowercase string identifier for the model.
provider (Categorical): The organization or laboratory that developed the model.
provider_slug (Text): A URL, string identifier for the provider.
aa_id (Text): A unique UUID assigned to the model within the database.
aa_intelligence_index (Continuous Numerical): A proprietary composite index measuring general reasoning
aa_coding_index (Continuous Numerical): A specialized index evaluating the model's proficiency in writi
composite_benchmark (Continuous Numerical): An aggregated baseline score calculated across multiple tra
mmlu_pro (Continuous Numerical): Massive Multitask Language Understanding Professional score.
```

```
gpqa_diamond (Continuous Numerical): Google-Proof Q&A benchmark score, representing advanced expertise i
humanitys_last_exam (Continuous Numerical): Performance on a highly complex benchmark designed to test t
livecodebench (Continuous Numerical): Performance on real-world, dynamic programming and algorithmic cha
scicode (Continuous Numerical): Evaluation of the model's ability to solve scientific coding problems an
pricing_tier (Categorical): A grouped classification of the model's cost bracket.
is_open_source (Binary): Indicates whether the model's weights are publicly accessible (TRUE) or restric
input_cost_usd_per_1m (Continuous Numerical): The cost in USD to send 1 million prompt tokens to the mod
output_cost_usd_per_1m (Continuous Numerical): The cost in USD to generate 1 million sampled output toke
blended_cost_usd_per_1m (Continuous Numerical): A weighted average cost per 1 million tokens, simulating
release_year (discrete numeric): The calendar year in which the model was first publicly released by its
parameter_count (discrete numeric): in billions of parameters. The reported size of the model. Only avai
```

Source code

```r
# libraries
library(knitr)
library(tidyverse)
library(broom)
library(htmltools)
opts_chunk$set(echo=FALSE) # hide source code in the document
# Data cleaning
df <- read_csv("../data/llm_price_performance_tracker_2026-03-31.csv", show_col_types = FALSE)

threshold <- 0.3

missing_fraction <- colMeans(is.na(df))
cols_to_keep <- names(missing_fraction[missing_fraction <= threshold])
extra_money_cols <- c("pricing_tier", "input_cost_usd_per_1m",
                      "output_cost_usd_per_1m", "blended_cost_usd_per_1m")
cols_to_keep <- union(cols_to_keep, intersect(extra_money_cols, names(df)))


df_filtered <- df %>% select(all_of(cols_to_keep))
llm <- df_filtered %>% drop_na()

llm <- llm %>% filter(blended_cost_usd_per_1m > 0)

# Price distribution.
ggplot(llm, aes(x = blended_cost_usd_per_1m)) +
  geom_histogram(bins = 30) +
  scale_x_log10() +
  labs(title = "Distribution of LLM blended cost",
       x = "USD per 1M tokens (log scale)", y = "Count") +
  theme(
    plot.title = element_text(size = 10),
  )
# Price vs each benchmark, side by side.
llm_long <- llm %>%
  pivot_longer(cols = c(mmlu_pro, gpqa_diamond, humanitys_last_exam,
                        livecodebench, scicode),
               names_to = "benchmark", values_to = "score")
```

```r
ggplot(llm_long, aes(x = blended_cost_usd_per_1m, y = score,
                     color = is_open_source)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  scale_x_log10() +
  facet_wrap(~ benchmark, scales = "free_y") +
  labs(title = "Benchmark score vs price, by skill",
       x = "Blended cost per 1M tokens (log scale, USD)",
       y = "Benchmark score",
       color = "Open source")
# Open vs closed source - performance gap by benchmark
ggplot(llm_long, aes(x = is_open_source, y = score, fill = is_open_source)) +
  geom_boxplot() +
  facet_wrap(~ benchmark, scales = "free_y") +
  labs(title = "Performance: open vs closed source, by benchmark",
       x = "Open source?", y = "Score")
# Summary table
llm %>%
  group_by(provider) %>%
  summarise(n = n(),
            mean_intel = mean(aa_intelligence_index, na.rm = TRUE),
            mean_coding = mean(aa_coding_index, na.rm = TRUE),
            mean_price = mean(blended_cost_usd_per_1m, na.rm = TRUE)) %>%
  arrange(desc(mean_intel))
cat(readLines('../data/README.md'), sep = '\n')
```